

M.Tech Project Review
on

Human Activity Recognition using Skeletal Dataset

by

Rishit Maurya (224ME1001)

Supervisor

Dr. Ravi Kumar Mandava

Asst. Professor



Department of Mechanical Engineering
IIITDM Kurnool

Contents



- **Introduction**
- **Applications**
- **Literature Review**
- **Research Gaps**
- **Objectives**
- **Work done so far**
- **Dataset Collection**
- **Deep Learning Models**
- **Results**
- **References**

Introduction about Human Activity Recognition



- ❑ **Human Activity Recognition (HAR)** refers to automatically identifying and classifying human actions from sensor or motion data.
- ❑ HAR analyzes **body movement patterns** to identify activities like walking, sitting, standing, and climbing stairs.
- ❑ Accurate HAR supports **abnormal behavior detection**, recovery monitoring, and safety for independent living.

Why HAR is needed ?

- ❑ Monitor patient movements for rehabilitation and recovery progress.
- ❑ Enhance surveillance systems with automatic threat detection.
- ❑ Enable robots to understand and respond to human actions.
- ❑ Improve interaction, navigation, and task coordination with humans.

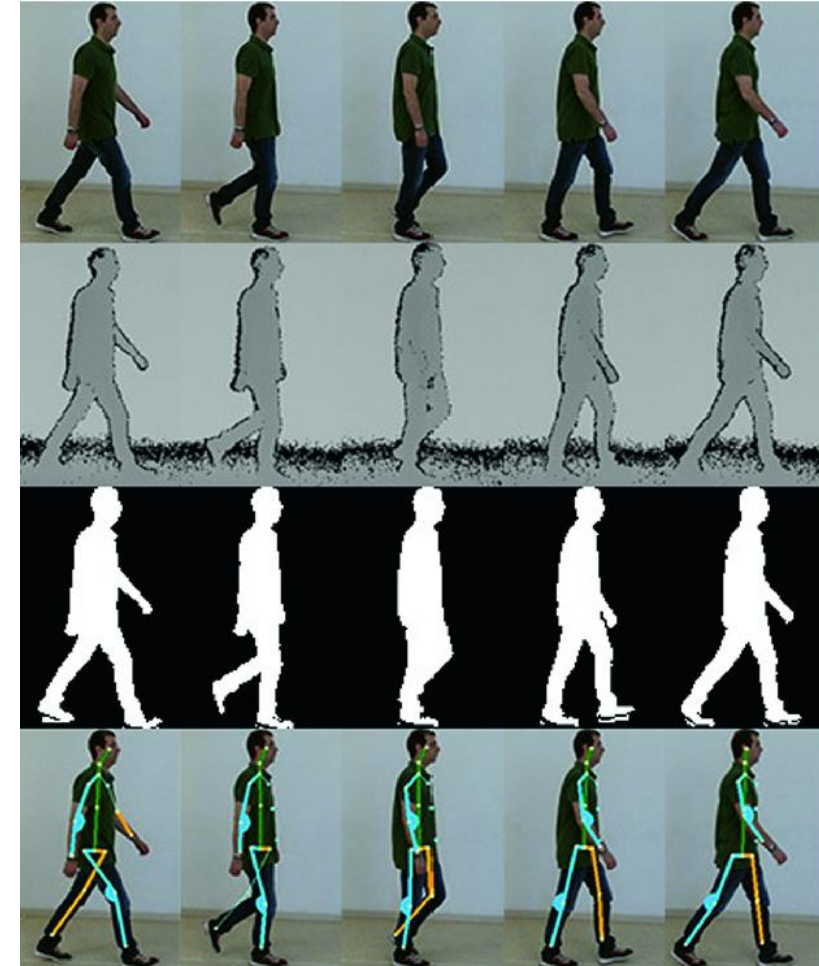


Fig.1. HAR

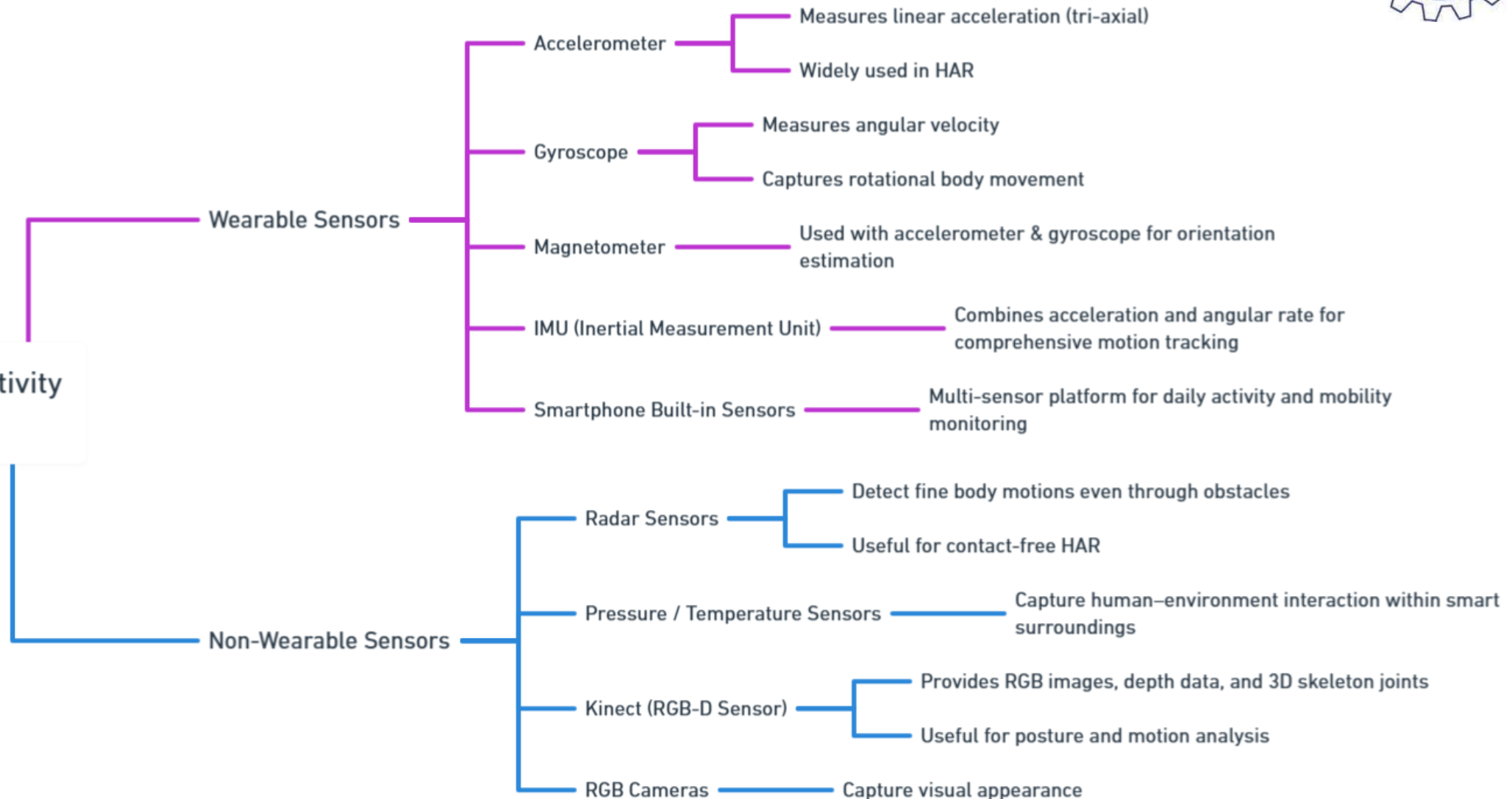
Applications of Human Activity Recognition



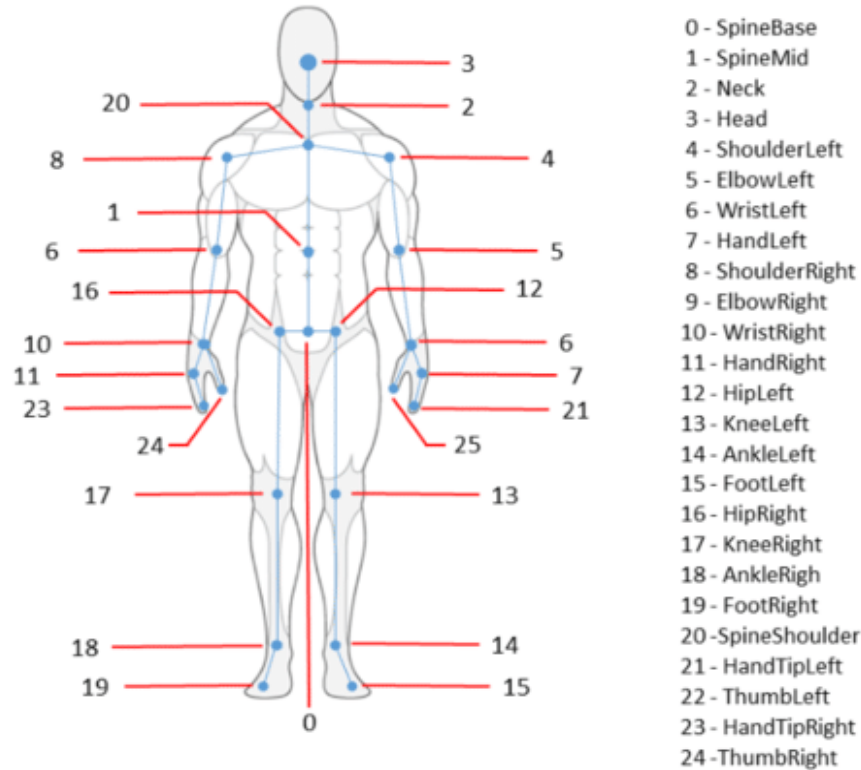
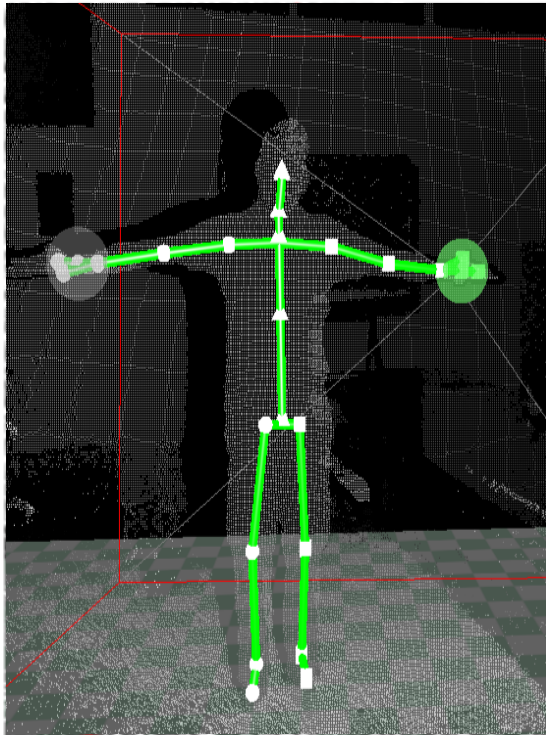
Sensor Classification for Human Activity Recognition



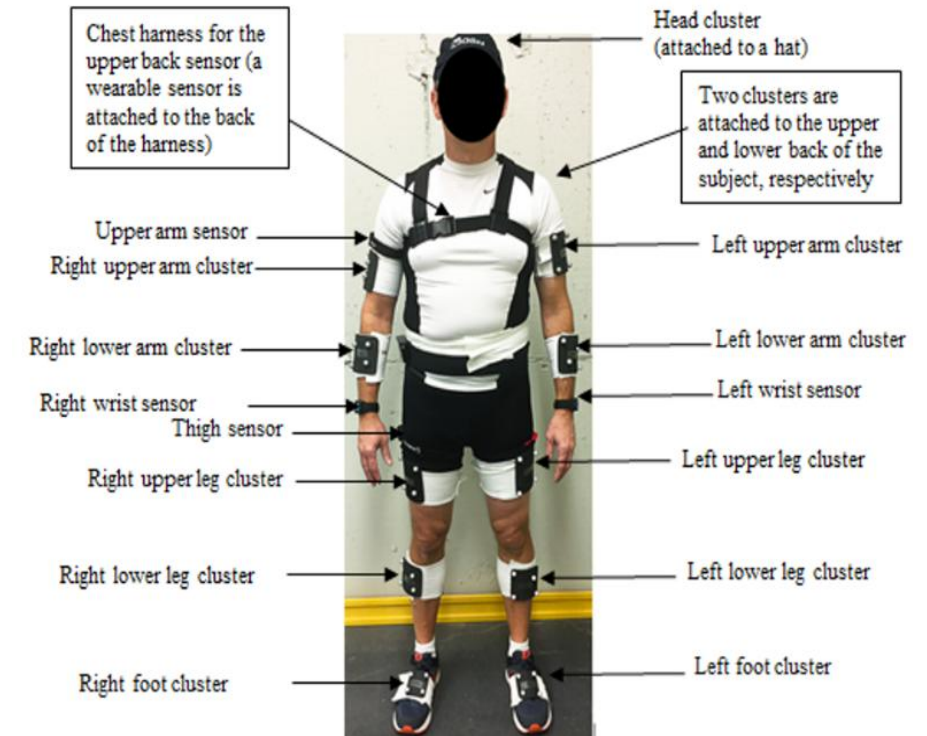
Sensors for Human Activity Recognition (HAR)



Different Approaches to Data Collection



(a)



(b)

Fig.2. Figure (a) Data collection using Kinect sensor and figure (b) Data collection using wearable sensors

Literature review



S. No.	Authors & Year	Title	Key Findings
1	Lalwani et al., 2024	A Novel CNN-BiLSTM-GRU Hybrid Deep Learning Model for Human Activity Recognition.	▪ The proposed CNN-BiLSTM-GRU hybrid model achieved an accuracy of 99.7% on the WISDM dataset ▪ The model processes raw wearable sensor data with minimal preprocessing. ▪ The findings have implications for bipedal robotics by enhancing human–robot interaction quality.
2	Ashwini K et al., 2020	Skeletal Data based Activity Recognition System	▪ The system uses statistical features derived from Kinect skeletal data. ▪ The Support Vector Machine (SVM) classifier performed better than the KNN classifier.
3	Rahul Kala et al., 2016	Human Activity Recognition in Real-Time Environments using Skeleton Joints	▪ The Artificial Neural Network (ANN) classifier achieved the highest aggregate accuracy result of 95.1%. ▪ Principal Component Analysis (PCA) was used for dimension reduction of 3D joint coordinates captured by the Kinect camera.
4	Komang Mandira et al., 2019	Human activity recognition using skeleton data and support vector machine	▪ The study proposed using Support Vector Machine (SVM) classification on skeleton data captured by a Kinect device with accuracy of 93.75%.
5	Khatun et al., 2022	Deep CNN-LSTM With Self-Attention Model for Human Activity Recognition Using Wearable Sensor	▪ A hybrid deep learning model combining CNN-LSTM and a self-attention algorithm was proposed to enhance predictive capabilities. ▪ The model uses raw smartphone sensor data (accelerometer, gyroscope, and linear acceleration) ▪ It achieved an accuracy of 99.93% on the collected H-Activity dataset.

S. No.	Authors & Year	Title	Key Findings
6	Açış & Güney., 2023	Classification of human movements by using Kinect sensor	- The maximum average classification success achieved was 98.85%. - This maximum success utilized LSTM for feature extraction combined with a machine learning classifier (SVM Kernel/Logistic Regression/Medium Neural Network). - The study concluded that joint data provided much more successful results for classification than image data.
7	Challa et al., 2022	A multibranch CNN-BiLSTM model for human activity recognition using wearable sensor data	- A multibranch CNN-BiLSTM network was proposed for automatic feature extraction from raw sensor data. - The model uses different convolutional filter sizes (3, 7, and 11) to capture various temporal local dependencies. - It achieved accuracies of 96.37% (UCI-HAR), 96.05% (WISDM), and 94.29% (PAMAP2) datasets.
8	Kaseris et al., 2024	A Comprehensive Survey on Deep Learning Methods in Human Activity Recognition	It investigates approaches utilizing diverse input modalities like accelerometer data, video sequences, and audio signals
9	Sarveshwaran, 2022	Investigation on Human Activity Recognition using Deep Learning	Discussion about the deep learning model, merits, demerits and dataset used in human activity recognition.
10	Morana et al., 2017	KARD - Kinect Activity Recognition Dataset	KARD contains 18 Activities. Each activity is performed 3 times by 10 different subjects.

A. Data and Annotation Challenges :

- Limited annotated datasets.
- Low subject diversity.
- Noisy, imbalanced real-world data.
- High-cost vision data processing.
- Reliance on handcrafted features.
- Deep models lack interpretability.

B. Activity and Complexity Modeling :

- Complex multi-sensor fusion required.
- Subtle/overlapping motions cause difficulty.
- Hard postural transition detection.
- Vision suffers occlusion/clutter.

C. Architectural and Methodological Limitations :

- Heavy preprocessing overhead.
- No multi-resolution encoding.
- High computational complexity.
- RNNs are slow, and provide limited context.
- CNN/RNN spatial–temporal mismatch.

D. Deployment and Real-World Challenges :

- Severe domain shift issues.
- Low wearable adoption.
- No adaptive recalibration.
- Embedded systems constraints.
- Vision-based privacy concerns.

Objective



- Develop a HAR model that **overcomes limitations** in data scarcity, multi-sensor activity complexity, architectural inefficiencies, and real-world deployment constraints.
- Perform Human Activity Recognition (HAR) using **non-wearable Kinect (RGB-D / skeleton) data**.
- Perform Human Activity Recognition (HAR) using **wearable IMU sensor data**.
- Integrate and compare activity recognition performance across **wearable and non-wearable sensing modalities**.
- **Map and transfer** the recognized human motion patterns to a **bipedal robot**.
- Enable the robot to **imitate and reproduce human activities** accurately in real time.

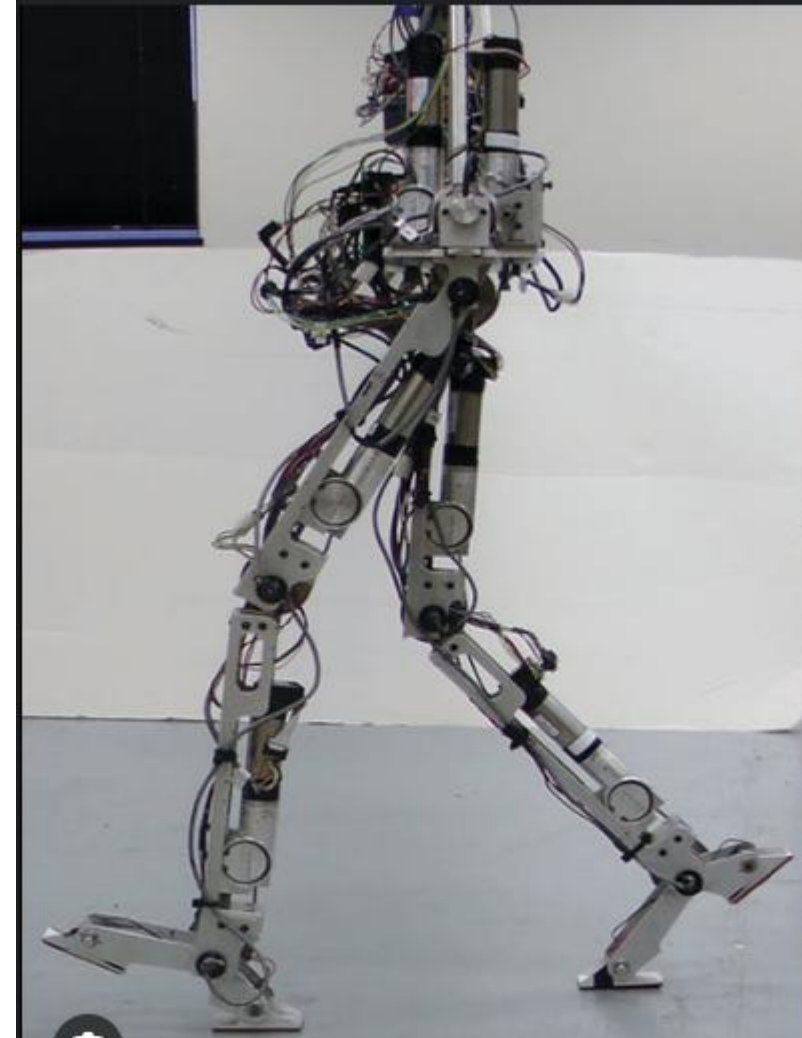


Fig.3. Bipedal Robot

Work done so far



- **Created a Skeletal Joints Dataset** using Microsoft Kinect Sensor for Human Activity Recognition.
- Studied skeletal joint movement patterns specific to each activity to enhance accurate feature capture for the model.
- Used a **multi-scale feature extraction** approach to **capture slow, medium, and fast** skeletal joint movements during activities for dataset creation.
- Developed a hybrid deep learning architecture **Multi-Scale Gated Residual Skip BiLSTM** model.
- Achieved a classification accuracy of **97.97%** using the MS-GRS-BiLSTM model on the skeletal joints dataset for Human Activity Recognition.
- Evaluated the MS-GRS-BiLSTM model on the **KARD - Kinect Activity Recognition dataset** and achieved an accuracy of **98.77%**.

Sensor Selection – Kinect Sensor



- ❑ Kinect provides **3D skeletal coordinates** (x, y, z) for 25 joints, supporting detailed posture and gait analysis.
- ❑ It **captures motion contact-free**, avoiding discomfort and ensuring user independence.
- ❑ Kinect **records full-body** movement in global coordinates, whereas wearables capture only local motion signals.
- ❑ Kinect **tracks multiple joints** simultaneously for coordinated movement detection; wearables typically monitor only one or few points, limiting context.



Fig.4. Kinect Sensor

Skeletal Joints Dataset - Using Kinect Sensor



- Data captured using **Microsoft Kinect** with **RGB, depth, and 3D skeletal** data.
- Kinect uses a **25-joint skeleton model** with **x, y, z** coordinates tracked over time.
- Dataset includes **30 individuals** for variability in movement and posture.
- Each person performs **7 activities**: **bending, jumping, running, sitting, squat, standing, walking**.
- Each activity is stored as a **time-series CSV** with **timestamp, body ID, and joint coordinates**.

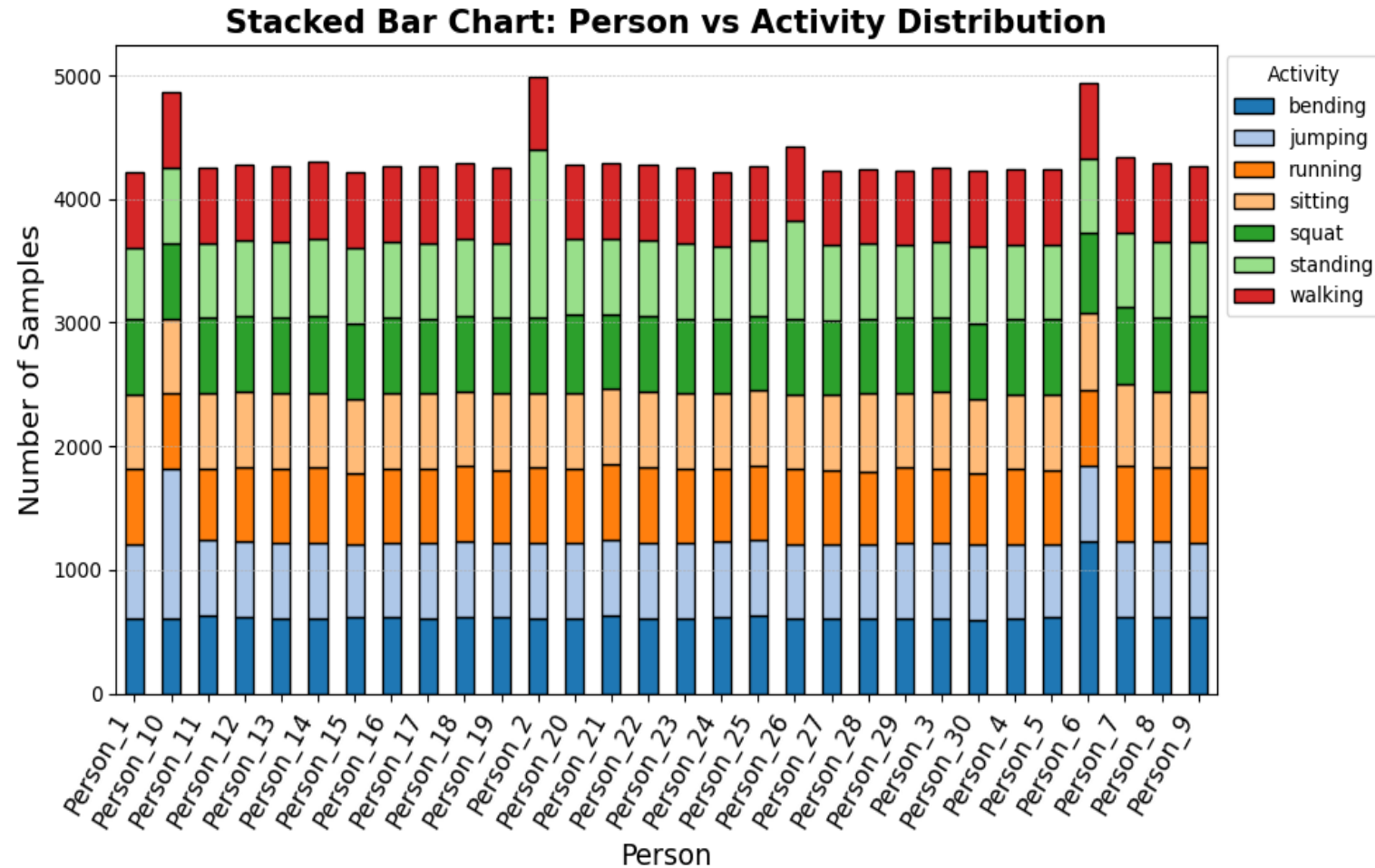


Fig.5. Person vs Activity Distribution

Dataset Collection

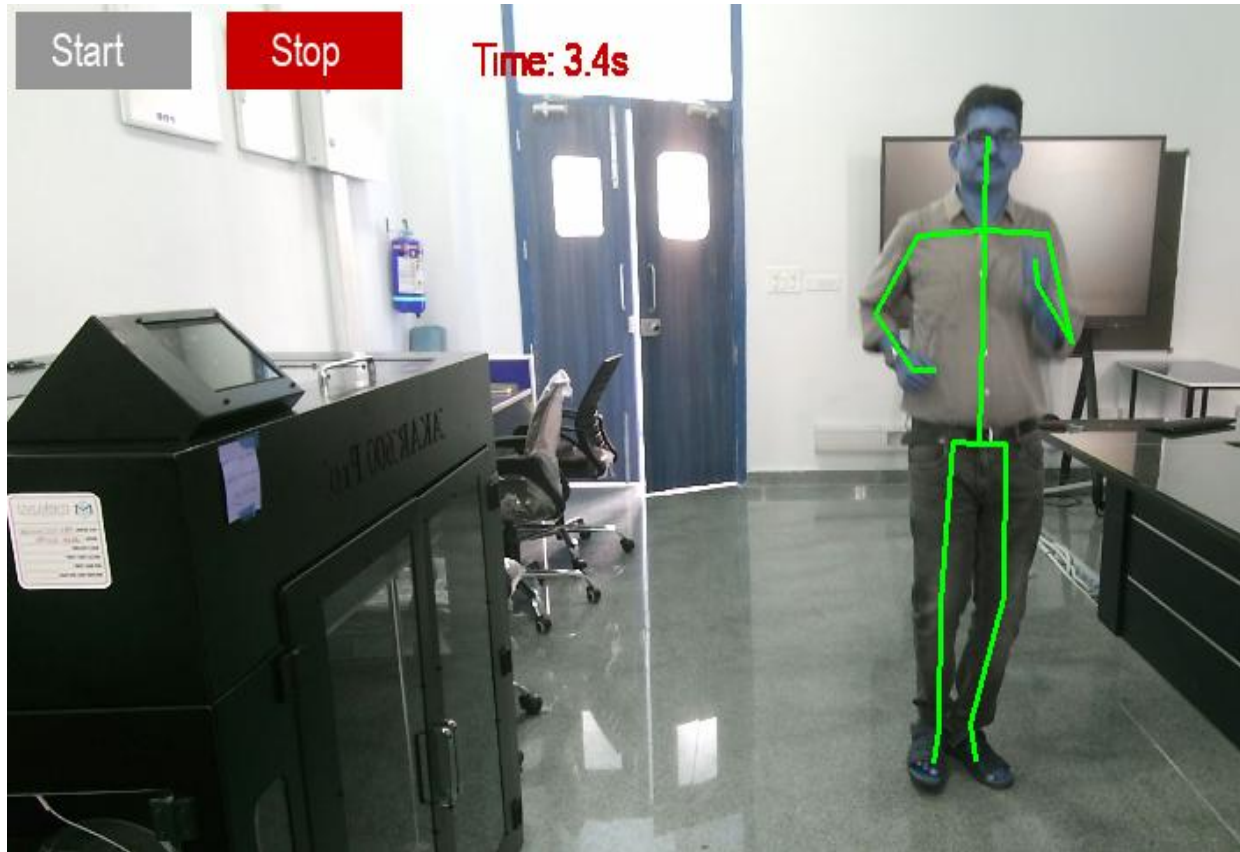


Fig.6. Person performing Activity for Dataset Collection

Skeleton Animation: Person_1 - running

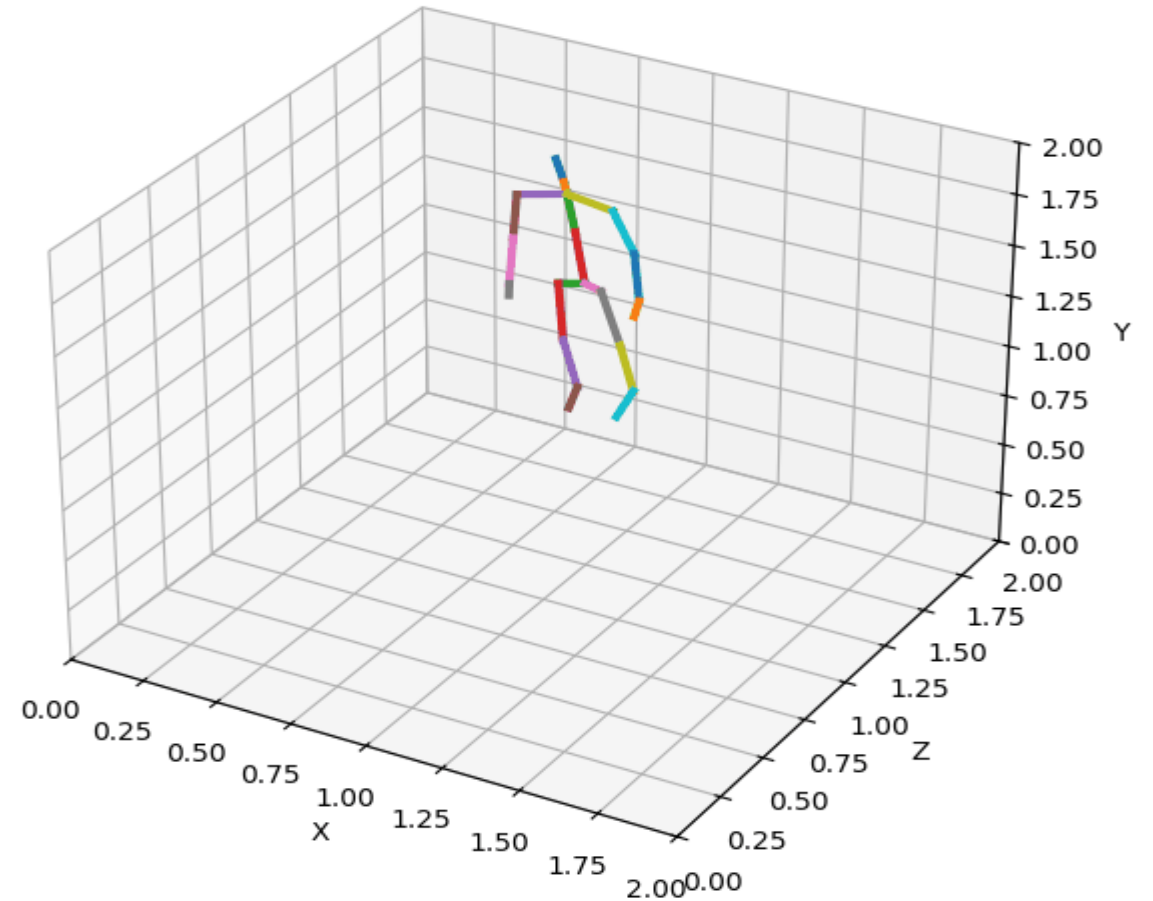


Fig.7. Skeletal representation of Activity

Models Used



Model	Key Idea	Architecture	Strengths	Limitations	Best Use Cases
Long Short Term Memory (LSTM)	Learns long-term dependencies in sequential data	Single-direction recurrent unit with input, forget, and output gates	- Handles vanishing gradient problem- Good for time-series / sequential classification. - Captures temporal memory effectively.	- Loses information from future context. - May underperform on complex motion patterns.	Basic sequence forecasting , simple activity recognition , speech modeling
Bidirectional LSTM (BiLSTM)	Uses past + future context for each time step	Two LSTM layers: one forward, one backward, outputs concatenated	- Strong contextual understanding- Captures bidirectional motion patterns. - Good for overlapping / subtle movements.	- Increased computational load. - More memory usage due to dual processing.	Human activity recognition , NLP tasks, gesture recognition
Gated Residual Skip BiLSTM (GRS-BiLSTM)	Enhances BiLSTM with Gated Residual Skip (GRS) connections	BiLSTM backbone + gating layer + residual skip pathways + optional multi-scale inputs	- Improved gradient flow and training stability- Better multi-resolution temporal modeling. - Handles subtle transitions and long motion dependencies- Reduces over-smoothing and information loss.	- Complex architecture requiring careful tuning . - Higher training time and model size.	High-precision activity recognition , multi-sensor fusion , real-world motion environments

Proposed Model – MS-GRS-BiLSTM



- ✓ **Name** : Multi-Scale Gated Residual Skip Bidirectional Long Short Term Memory (MS-GRS-BiLSTM).
- ✓ **Goal** : classify short time-windows of sensor / joint data into activity classes.
- ✓ **Key ideas** : multi-scale temporal representation, learned gates to fuse scales, residual skip from raw input, BiLSTM temporal modeling.
- ✓ The MS-GRS-BiLSTM is **not just detecting momentary movement**, but is:

Understanding the shape of motion over time, at multiple speeds, while carefully preserving useful information and filtering noise.

Why Multi-Scale (MS) is Needed ?

Different activities show patterns at **different time speeds**:

Activity Type	Activity Speed	Example
Quick & Repetitive	Fast	Running, Jumping
Moderate Movement	Medium	Walking
Postural & Stable	Slow	Standing, Sitting

To capture these differences, the model **creates 3 versions** of each input window:

- Full detail (original resolution)
- Medium-scaled (every 2nd time step)
- Coarse-scaled (every 4th time step)

This allows the network to **understand both short, rapid motions and longer, slower movement trends**.

Formula :

$$x^{(1)} = x_t$$

$$x^{(2)} = \text{Downsample}(x_t, 2)$$

$$x^{(3)} = \text{Downsample}(x_t, 4)$$

$$h^{(k)} = \text{BiLSTM}(x^{(k)}) \quad \text{for } k \in \{1, 2, 3\}$$

$$H = \text{Concat}(h^{(1)}, h^{(2)}, h^{(3)})$$

Symbols

- $x^{(k)}$: Input at different temporal scales
- $\text{Downsample}(\cdot)$: Reduces sequence frequency
- $h^{(k)}$: BiLSTM features at each scale
- H : Combined multi-scale feature matrix

Gated Residual Skip (GRS) Blocks contains :

1. **Projection Layer** : Simplifies the signal and extracts essential structure.
2. **BiLSTM Layer** : Reads the movement pattern **forward and backward in time**.
3. This is important because **what happened just before and just after each moment** helps define the action (e.g., lowering before a jump).
4. **Gating Mechanism** : A learned **filter** that decides:
 - i) How much **new information** to use
 - ii) How much **previous information** to keep
5. **Residual Skip Connection** : Passes earlier information forward **without loss**, preventing the model from forgetting important details.

Formula :

$$g = \sigma(W_g H + b_g)$$

$$H_{update} = g \odot F(H)$$

$$H_{out} = H + H_{update}$$

Symbols

- H : Input feature representation
- $F(H)$: Learned transformed feature (e.g., another layer output)
- g : Gate controlling how much new information is added
- H_{update} : Gated new information
- H_{out} : Final output with skip (residual) connection
- $+$: Skip (residual) addition preserving original information

LSTM is a type of neural network used for **time-series or sequential data**.

- It has **memory cells** that allow it to **retain important past information**.
- It uses **gates** to decide what to **keep**, what to **update**, and what to **forget**.

A BiLSTM processes the sequence in **two directions**:

- Forward: earlier $\rightarrow$ later time steps
- Backward: later $\rightarrow$ earlier time steps

This gives it **context from both past and future** time steps,

Formula :

$$\vec{h}_t = \text{LSTM}_{\text{forward}}(x_t)$$

$$\overleftarrow{h}_t = \text{LSTM}_{\text{backward}}(x_t)$$

$$h_t = [\vec{h}_t \parallel \overleftarrow{h}_t]$$

Symbols

- $\vec{h}_t$: Forward hidden state (past $\rightarrow$ future direction)
- $\overleftarrow{h}_t$: Backward hidden state (future $\rightarrow$ past direction)
- $\parallel$: Concatenation operator

Formula :

$$i_t = \sigma(W_i x_t + U_i h_{t-1} + b_i) \quad (\text{input gate})$$

$$f_t = \sigma(W_f x_t + U_f h_{t-1} + b_f) \quad (\text{forget gate})$$

$$o_t = \sigma(W_o x_t + U_o h_{t-1} + b_o) \quad (\text{output gate})$$

$$\tilde{c}_t = \tanh(W_c x_t + U_c h_{t-1} + b_c) \quad (\text{candidate cell state})$$

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t \quad (\text{updated cell state})$$

$$h_t = o_t \odot \tanh(c_t) \quad (\text{hidden state output})$$

Symbols

- x_t : Input vector at time step t
- h_t : Hidden state at time t
- c_t : Memory (cell state) at time t
- i_t, f_t, o_t : Input, forget, and output gates
- $\tilde{c}_t$: Candidate memory update
- W_*, U_* : Weight matrices
- b_* : Bias terms
- σ : Sigmoid activation function
- $\tanh$: Hyperbolic tangent activation
- $\odot$: Element-wise multiplication

MS-GRS-BiLSTM Architecture

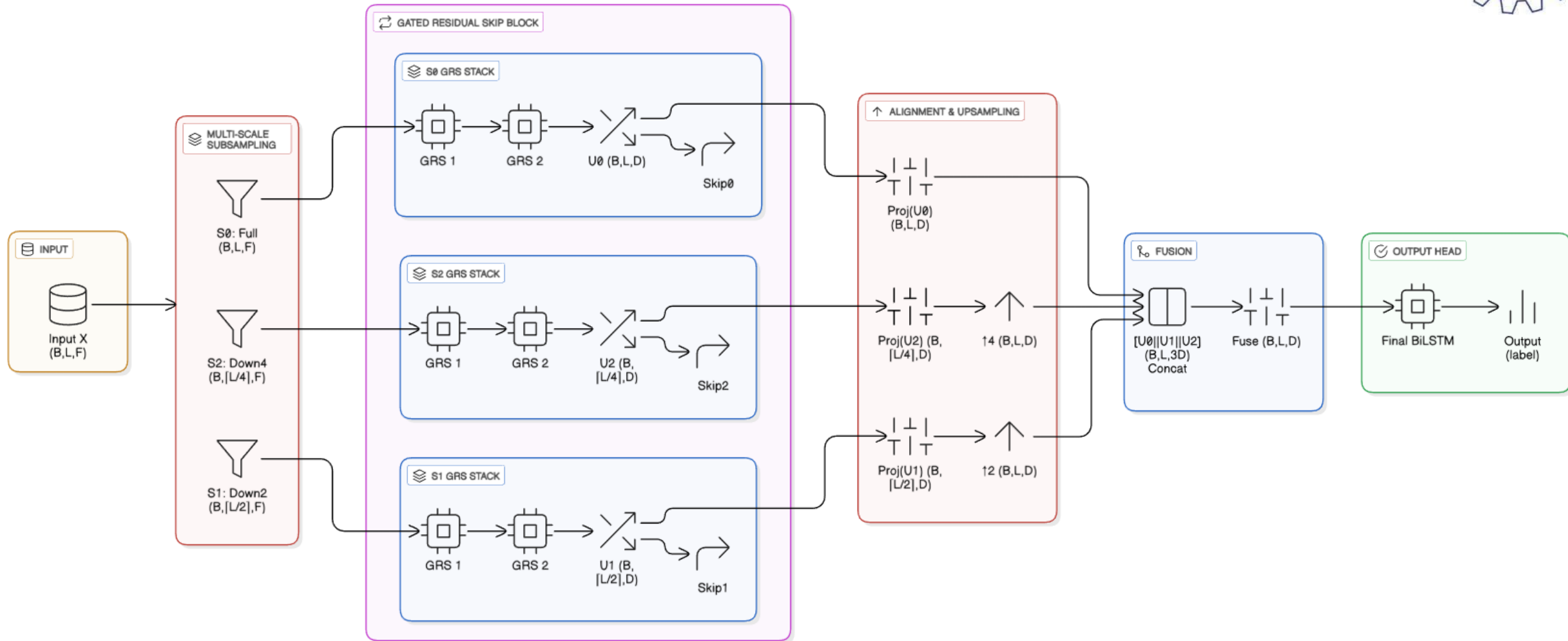


Fig.8. MS-GRS-BiLSTM Architecture

Metrics Used in HAR



Metrics	Definition	Formula	Usage in HAR	Symbols & Meaning
Precision	Proportion of correctly predicted positives out of all predicted positives.	$\text{Precision} = \frac{TP}{TP + FP}$	Important when false positives are costly (e.g., detecting a fall when none occurred).	TP: True Positives (correct positive predictions) FP: False Positives (incorrect positive predictions)
Recall	Proportion of correctly predicted positives out of all actual positives.	$\text{Recall} = \frac{TP}{TP + FN}$	Key when missing a positive is critical (e.g., failing to detect a real fall).	FN: False Negatives (missed actual positives)
F1-Score	Harmonic mean of Precision and Recall; balances false positives and false negatives.	$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$	Used when both precision and recall matter , e.g., imbalanced HAR activity classes.	-
Accuracy	Proportion of correctly predicted samples out of all samples.	$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$	General metric for overall performance; less reliable for imbalanced datasets .	TN: True Negatives (correct negative predictions)
Loss	Measures prediction error during training; used to update network weights.	Categorical Cross-Entropy: $\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N \sum_{c=1}^C y_{i,c} \log(p_{i,c})$	Guides weight updates in BiLSTM, helps learn sequential HAR patterns accurately.	N: Total samples C: Number of classes y_{i,c} : Ground-truth (1 if sample i belongs to class c, else 0) p_{i,c} : Predicted probability for class c for sample i

Results – a) My Dataset



Class	Precision	Recall	F1-Score
Bending	0.9982	0.9972	0.9977
Jumping	0.9344	0.9597	0.9469
Running	0.9571	0.9365	0.9467
Sitting	1.0000	1.0000	1.0000
Squat	1.0000	0.9952	0.9976
Standing	0.9955	1.0000	0.9977
Walking	1.0000	0.9942	0.9971
Overall Metric	Precision	Recall	F1-Score
Accuracy	-	-	0.9835
Macro Avg	0.9836	0.9833	0.9834
Weighted Avg	0.9836	0.9835	0.9835

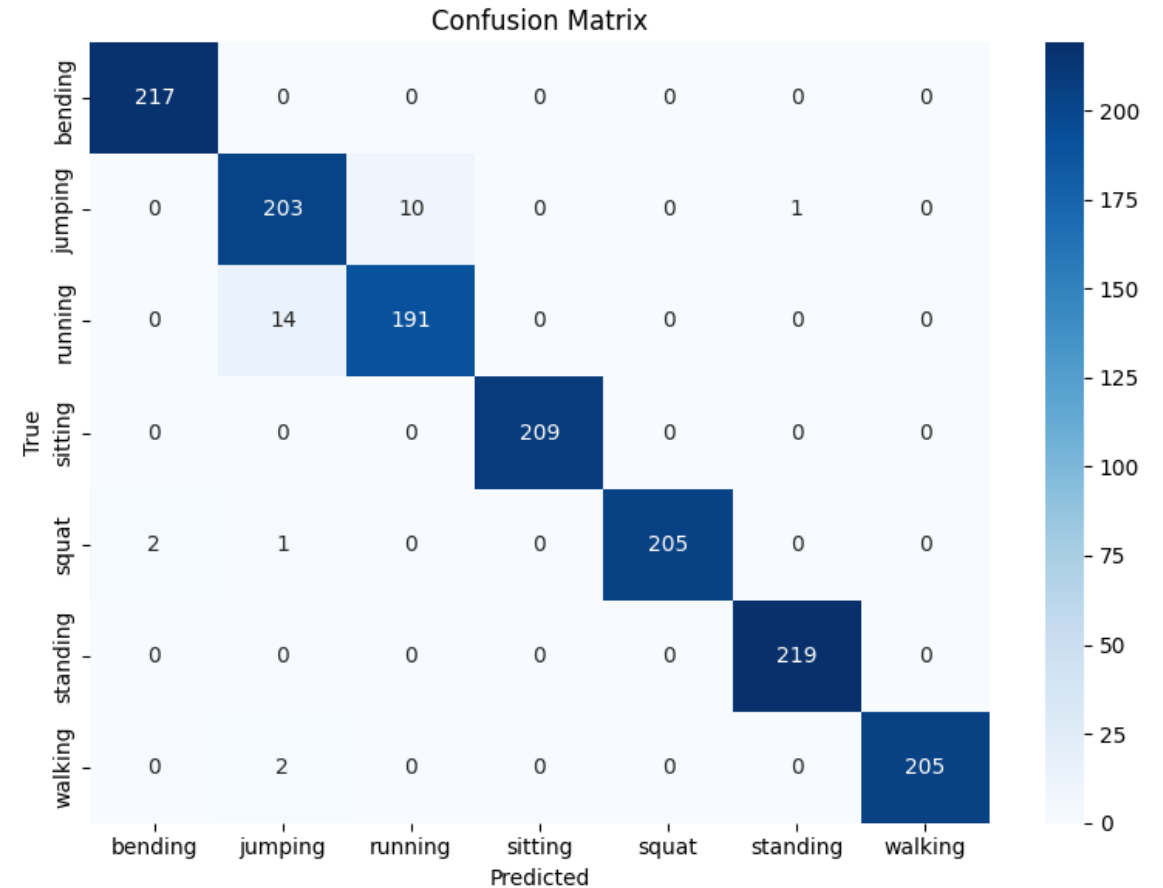


Fig.9. Confusion Matrix on Skeletal Dataset

Results – b) Online Dataset (KARD)



Class	Precision	Recall	F1-Score
Action_0	1.0000	1.0000	1.0000
Action_1	1.0000	1.0000	1.0000
Action_2	1.0000	1.0000	1.0000
Action_3	1.0000	1.0000	1.0000
Action_4	0.8333	1.0000	0.9091
Action_5	1.0000	1.0000	1.0000
Action_6	1.0000	1.0000	1.0000
Action_7	1.0000	1.0000	1.0000
Action_8	1.0000	1.0000	1.0000
Action_9	1.0000	1.0000	1.0000
Action_10	1.0000	1.0000	1.0000
Action_11	1.0000	1.0000	1.0000
Action_12	1.0000	1.0000	1.0000
Action_13	1.0000	1.0000	1.0000
Action_14	1.0000	1.0000	1.0000
Action_15	1.0000	0.7500	0.8571
Action_16	1.0000	1.0000	1.0000
Action_17	1.0000	1.0000	1.0000
Overall Metric	Precision	Recall	F1-Score
Macro Avg	0.9907	0.9861	0.9870
Weighted Avg	0.9897	0.9877	0.9833

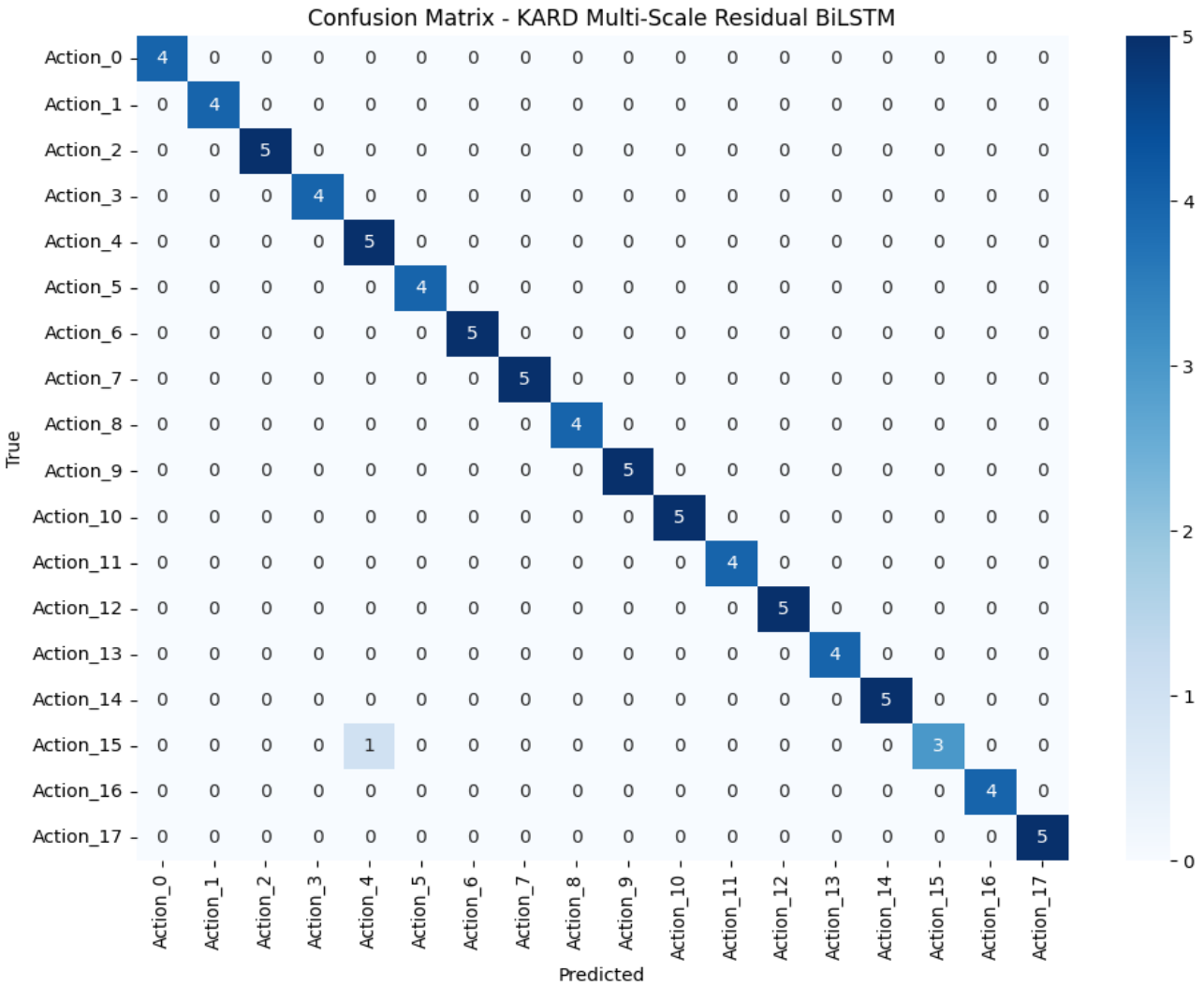


Fig.10. Confusion Matrix on KARD Dataset

Comparison on Skeletal Dataset



Activities / Models	LSTM	BiLSTM	GRS-BiLSTM	MS-GRS-BiLSTM
Bending	0.9091	0.9800	1.0000	0.9977
Jumping	0.8571	0.4900	0.7300	0.9469
Running	0.7692	0.7400	0.8200	0.9467
Sitting	0.4000	1.0000	0.9800	1.0000
Squat	1.0000	1.0000	1.0000	0.9976
Standing	0.5000	0.9200	0.9800	0.9977
Walking	0.8333	0.9700	1.0000	0.9971

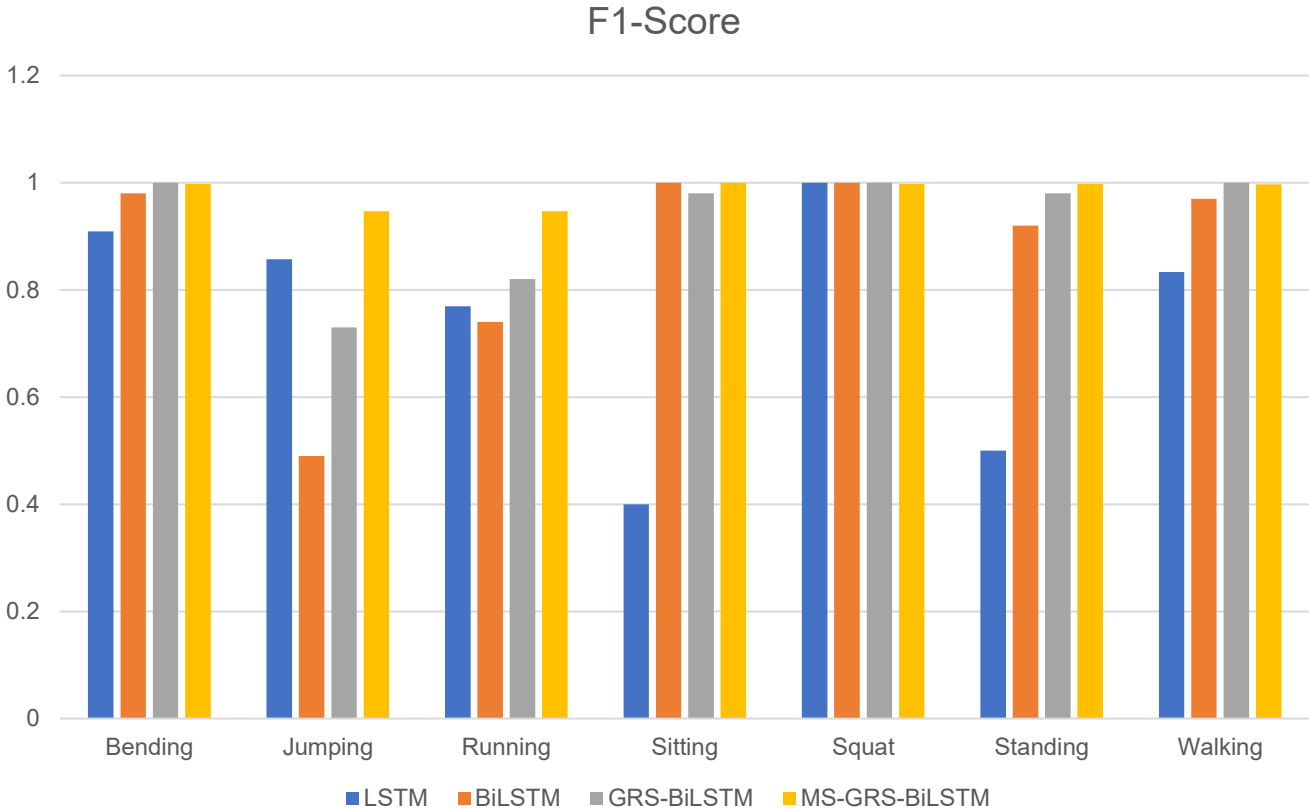


Fig.11. F1-Score amongst different Activity and Models

Comparison of Models Performance between Dataset



Skeletal Joint Dataset (My Dataset) :

Model	Test Loss	Test Accuracy	Precision	Recall	F1-Score	Model Size	Avg Inference Time per Sample
LSTM	1.3108	76.19%	0.7568	0.7619	0.7527	1.86 MB	31.01 ms
BiLSTM	0.3456	88.10%	0.8900	0.8800	0.8700	1.43 MB	27.03 ms
GRS-BiLSTM	0.1671	93.33%	0.9400	0.9300	0.9300	22.56 MB	31.86 ms
MS-GRS-BiLSTM	0.0516	98.35%	0.9836	0.9833	0.9834	6.82 MB	1.857 ms

KARD Dataset :

Model	Test Loss	Test Accuracy	Precision	Recall	F1-Score	Model Size	Inference Time per Sample
LSTM	0.9281	77.78%	0.7792	0.7750	0.7587	0.04 MB	0.015 ms
BiLSTM	0.5001	87.65%	0.8896	0.8750	0.8740	0.09 MB	0.077 ms
GRS-BiLSTM	0.3790	92.59%	0.9697	0.9222	0.9240	0.29 MB	0.037 ms
MS-GRS-BiLSTM	0.0573	98.77%	0.9907	0.9861	0.9870	3.58 MB	0.020 ms

Prediction of Activity using Proposed Model



Predicted Activity: squats

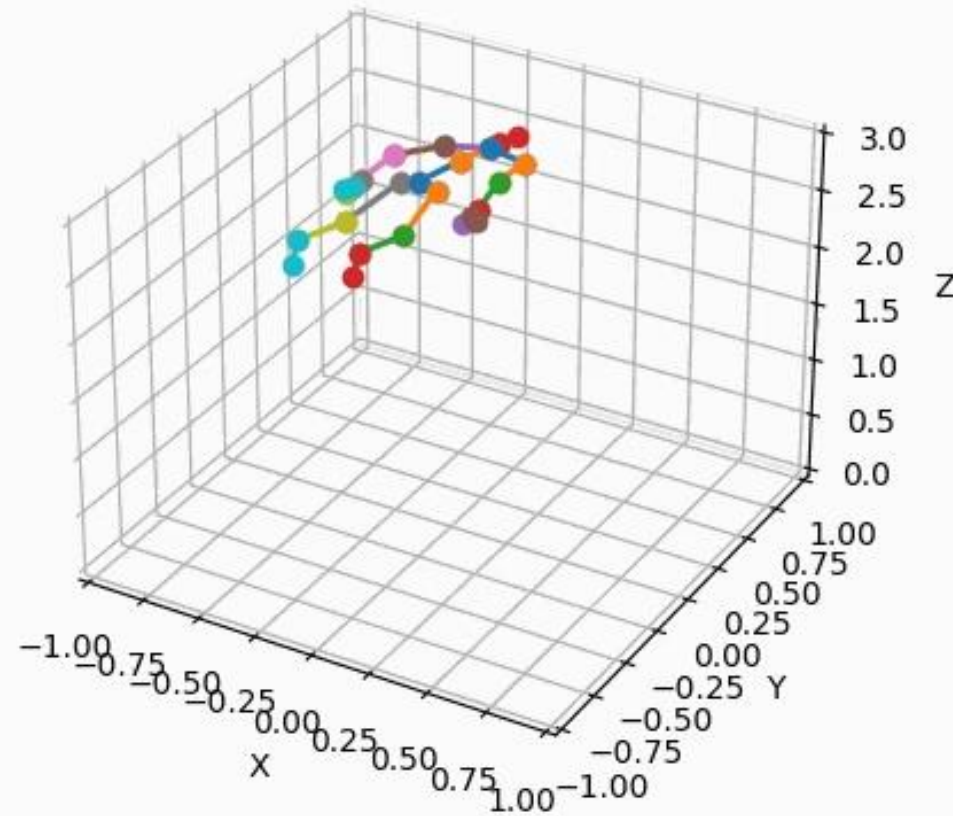


Fig.12. Prediction on different activities



1. Gupta, S. Deep learning based human activity recognition (HAR) using wearable sensor data. *Int. J. Inf. Manag. Data Insights* 2021, 1, 100046.
2. Diraco, G.; Rescio, G.; Caroppo, A.; Manni, A.; Leone, A. Human Action Recognition in Smart Living Services and Applications: Context Awareness, Data Availability, Personalization, and Privacy. *Sensors* 2023, 23, 6040.
3. Shuvo, M.M.H.; Ahmed, N.; Nouduri, K.; Palaniappan, K. A Hybrid Approach for Human Activity Recognition with Support Vector Machine and 1D Convolutional Neural Network. In *Proceedings of the 2020 IEEE Applied Imagery Pattern Recognition Workshop (AIPR)*, Washington, DC, USA, 13–15 October 2020; pp. 1–5.
4. Rojanavas, P.; Jantawong, P.; Jitpattanakul, A.; Mekruksavanich, S. Improving Inertial Sensor-based Human Activity Recognition using Ensemble Deep Learning. In *Proceedings of the 2023 Joint International Conference on Digital Arts, Media and Technology with ECTI Northern Section Conference on Electrical, Electronics, Computer and Telecommunications Engineering (ECTI DAMT & NCON)*, Phuket, Thailand, 22–25 March 2023; pp. 488–492.
5. Muhoza, A.C.; Bergeret, E.; Brdys, C.; Gary, F. Multi-Position Human Activity Recognition using a Multi-Modal Deep Convolutional Neural Network. In *Proceedings of the 2023 8th International Conference on Smart and Sustainable Technologies (SpliTech)*, Split, Croatia, 20–23 June 2023; pp. 1–5.
6. Tao, S.; Goh, W.L.; Gao, Y. A Convolved Self-Attention Model for IMU-based Gait Detection and Human Activity Recognition. In *Proceedings of the 2023 IEEE 5th International Conference on Artificial Intelligence Circuits and Systems (AICAS)*, Hangzhou, China, 11–13 June 2023; pp. 1–5.



7. Hassler, A.P.; Menasalvas, E.; García-García, F.J.; Rodríguez-Mañas, L.; Holzinger, A. Importance of medical data preprocessing in predictive modeling and risk factor discovery for the frailty syndrome. *BMC Med. Inform. Decis. Mak.* 2019, 19, 33.
8. Xu, S.; Zhang, L.; Huang, W.; Wu, H.; Song, A. Deformable convolutional networks for multimodal human activity recognition using wearable sensors. *IEEE Trans. Instrum. Meas.* 2022, 71, 2505414.
9. Beddiar, D.R.; Nini, B.; Sabokrou, M.; Hadid, A. Vision-based human activity recognition: A survey. *Multimed. Tools Appl.* 2020, 79, 30509–30555.
10. Lara, O.D.; Labrador, M.A. A survey on human activity recognition using wearable sensors. *IEEE Commun. Surv. Tutorials* 2012, 15, 1192–1209.
11. Ke, S.R.; Thuc, H.L.U.; Lee, Y.J.; Hwang, J.N.; Yoo, J.H.; Choi, K.H. A review on video-based human activity recognition. *Computers* 2013, 2, 88–131.
12. Ray, A.; Kolekar, M.H.; Balasubramanian, R.; Hafiane, A. Transfer learning enhanced vision-based human activity recognition: A decade-long analysis. *Int. J. Inf. Manag. Data Insights* 2023, 3, 100142. [CrossRef]
13. Singh, R.; Kushwaha, A.K.S.; Srivastava, R. Recent trends in human activity recognition—A comparative study. *Cogn. Syst. Res.* 2023, 77, 30–44.



THANK YOU